FALCON SCALING

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Test Disclaimer:

All work submitted as part of this evaluation remains the sole intellectual property of the candidate. Falcon Scaling and PolarityIQ make no claim of ownership over any datasets, code, tools, pipelines, products, or analyses created during this testing process. Your work is yours — regardless of the outcome.

72-HOUR Differentiator Questions

Instructions:

Submission for All Four Tasks: Please submit all deliverables in a single email to optimize@falconscaling.com and request acknowledgement of receipt.

How You Will Be Evaluated:

1) Your performance will be evaluated relative to the other candidates (top 1% of applicant pool).

2) This position requires the skilled use of A.I. but it also requires the critical thinking, problem solving, and decision making of a human being.

Candidates that integrate DEEP LEVELS of human thought ALONGSIDE A.I. insights will score highest.

Task #1 DATASET CREATION

GOAL: Create a granular/detailed/validated dataset (from scratch) of International Family Office Intelligence.

What is a Family Office (FO)?

A family office is a private, dedicated entity established by ultra-high-net-worth families to manage their wealth, investments, taxes, philanthropy, and legal affairs. They provide tailored, comprehensive services to grow and preserve wealth for multiple generations. Common types include Single-Family Offices (SFO), which serve one family, and Multi-Family Offices (MFO), which share resources across families.

You should:

- Determine on your own **HOW LARGE** of a dataset you want to create. **200 family office records is the floor for this task.** How far above that 200 minimum threshold you want to go is completely your discretion. Quality trumps quantity.

- Determine how **GRANULAR** (detailed) of a dataset you will want to create (how much detail will you integrate into your dataset). Granularity / Level of detail will be one of the most important factors for this task.

Here are SOME hypothetical family office intelligence components (these are examples of the aforementioned "granularity" we mentioned above) to assist you in getting started (you can build your dataset to include some of the details, all of the details, and of course you can add your own discretionary intelligence as well).

Basic Family Office Details You Might Want to Include in your Dataset

FO Firm Name
FO Decision Maker Names
FO Decision Maker Roles
FO Decision Maker Contact Information (email, Linkedin, phone)
FO Portfolio Companies
FO Investment Themes
FO Fund Relationships
FO Website Address

Family Office Signals You Might Want to Include in your Dataset

Recent FO News
Recent FO LinkedIn Activity
Recent FO Filings
Recent FO Portfolio Announcements

Other Family Office Details You Might Want to Include in your Dataset

Investment focus
Acceptable check size range
LP relationships

Investment Strategy
Sector Focus
Co-Invest Frequency
Co-Investor Relationships
Single Family Offices vs Multi-Family Offices

Here is an actual example of a Family Office Dataset to give you an idea of one particular approach. This is only an EXAMPLE.

Your final submission will hopefully be superior in scale, granularity, and value.

https://app.box.com/s/4voy48mepvmsa6c0qgi9smdhrsnkquqa

(Do not replicate the structure of this example blindly; independent judgment regarding schema design and intelligence layering will be evaluated).

And here is our Family Office SaaS Platform for Reference as well (which may give you additional context for this task):

Polarity IQ

Here are your required deliverables for Task #1:

- A final, validated, structured dataset (CSV or XLSX) - minimum of 200 records.
- A documentation file outlining your tech stack, acquisition methodology, data enrichment methodology, validation methodology, sources used, challenges faced, and key insights.
- Your final dataset must not be "theoretical"; this dataset must include REAL WORLD intelligence (an Excel or Google Sheets output file) to confirm that your solution actually works.

Task #2 RAG PIPELINE BUILD

GOAL: Build a functional RAG (Retrieval-Augmented Generation) pipeline that makes your Task #1 dataset queryable via natural language.

Task #2 builds directly on Task #1. You will ingest the exact dataset you created in Task #1 into a RAG pipeline and make it queryable via natural language. PolarityIQ's core value proposition is transforming raw family office data into actionable, queryable intelligence — this task asks you to build a working prototype of exactly that.

The quality of your Task #1 dataset directly determines the power of your Task #2 demo. A rich, granular dataset produces a genuinely useful intelligence tool. A shallow dataset produces a weak one.

What To Build:

- Ingest your Task #1 family office dataset into a vector database of your choice (ChromaDB, Pinecone, Supabase pgvector, Weaviate, or equivalent).
- Build a retrieval pipeline that accepts natural language queries and returns accurate, grounded responses from your dataset.
- Demonstrate your pipeline answering at least 3 example queries against your real dataset. Examples: "Which family offices in your dataset focus on AI with check sizes above $10M?" or "Show me all single-family offices that have made direct investments in the last 12 months."
- Your stack choice, chunking strategy, embedding model selection, and retrieval approach are entirely at your discretion. Justify your decisions in your documentation.

Required Deliverables for Task #2:

- A working GitHub repository (or shared code file) containing your full RAG pipeline.
- A demonstrable interface — either a live publicly accessible URL (Streamlit Cloud, Hugging Face Spaces, Render, or equivalent) or a screen recording showing real queries returning accurate results from your actual dataset. If submitting a live interface, provide the URL directly in your email. Note: a live queryable interface will be weighted more heavily than a screen recording, as it allows us to test your system directly with our own queries.
- A brief documentation note (can be part of your Task #1 documentation file) covering: stack choices, chunking strategy, embedding model, retrieval approach, what failed, and what you would improve with more time.

Note: A functional but imperfect RAG pipeline with honest documentation of its limitations will score significantly higher than a polished demo that does not actually work on your real dataset.

Task #3 SaaS CONVERSION ANALYSIS

GOAL: Study our PolarityIQ SaaS Platform (https://app.polarityiq.com) and design a rigorous, evidence-based plan to improve our free trial to paid user conversion rate.

In other words: how would you increase the percentage of free-trial users who become paying customers?

(remember, how YOU think here is important, GENERIC A.I. copy and paste answers will significantly decrease your score - candidates that include HIGHLY SPECIFIC recommendations will score highest)

Task #4 AI PRODUCT BUILD

We have access to a network of 300,000+ investors.

Your Goal: Design and build a "trip-wire" investment product (from scratch) you believe we could sell to these 300,000 investors. Your product can focus on any niche within the investment landscape and can take any form, provided it delivers immediate, tangible value to your ICP.

A trip-wire product is typically a low-cost entry product priced between $47 and $1,000. Once someone purchases a trip-wire, we then "ascend" them to higher priced products and services.

For inspiration, you can review these three trip-wire products here:

- Family Office Catalyst (this product is perfectly suited (aligned) because it enables us to easily "ascend" buyers to our Family Office SaaS platform or to other more robust data products.)
- Private Markets Investing Catalyst Suite
- Ultimate Valuation Toolkit

Each of these products was fully developed using A.I.

We want you to create a genuinely new investment product, using the three examples above only as inspiration.

Here is what you should include in your final submission:

- The problem or pain point your product solves:
- The user/ICP it serves
- Your product's price
- What might be a viable ascension path for your product

You are free to include in your product build:

- Excel Models / Dashboard / Excel Tools

- Word Based Guides
- Custom GPTs
- AI Agents
- A Working SaaS Prototype

We are not looking for a pitch deck or a concept. We are looking for something a real investor could purchase today.

The strongest solution will identify a specific, painful investment challenge, define a clear ascension path, build something that genuinely solves it, and explain concisely why this audience would pay for it.

Your submission should include a step-by-step explanation as to how you built this product, including all platforms/apps utilized.

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